

Problem Statement

The Global Superstore dataset contains a large volume of transactional sales data, where information related to customers, products, regions, dates, sales, quantity, discount, and profit is stored together. Since the same customer, product, and other descriptive information may appear repeatedly across multiple transactions, the raw transactional structure can become difficult to manage, analyze, and use for efficient business reporting.

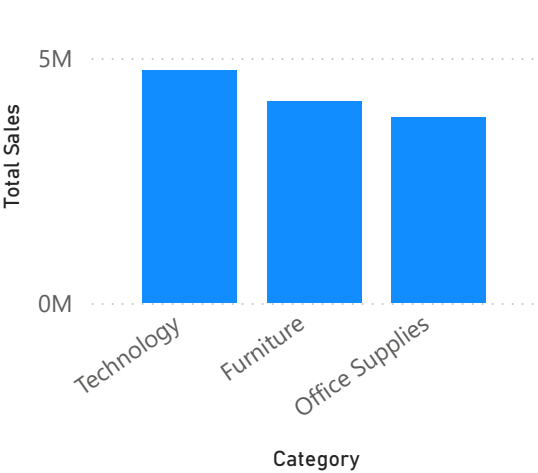
The problem is therefore to transform the raw transactional data into a well-structured dimensional data model using Microsoft Power BI. The measurable business transactions need to be separated into a central Fact_Sales table, while descriptive information needs to be organized into appropriate Dimension tables such as Dim_Customer, Dim_Product, Dim_Region, and Dim_Date. These Fact and Dimension tables must be connected through appropriate one-to-many relationships to form a Star Schema. The model should provide a clear and efficient structure for analyzing business performance from different perspectives, including customer, product, region, and time.

Finally, the dimensional model should support the creation of DAX measures such as Total Sales, Total Profit, Profit Margin, and Average Order Value. These measures can then be used with Power BI charts, cards, and slicers to produce interactive reports and support effective business analysis and decision-making.

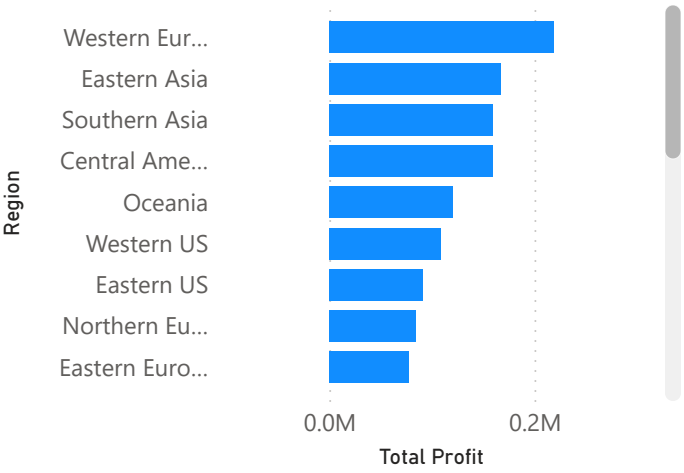
Solution

- Create a Fact_Sales table containing sales transactions and numerical measures.
- Create Dim_Customer, Dim_Product, Dim_Region, and Dim_Date tables containing descriptive information.
- Remove unnecessary columns and duplicate dimension records using Power Query.
- Establish one-to-many relationships between the Dimension tables and Fact_Sales.
- Create DAX measures for Total Sales, Total Profit, Profit Margin, and Average Order Value.
- Develop interactive Power BI visualizations and slicers for business analysis.

Total Sales by Category



Total Profit by Region



Total Sales by Year

